The documentation and process conversion measures necessary to comply with this document shall be completed by 21 October 2024.

INCH-POUND

MIL-PRF-19500/735C
w/AMENDMENT 2
24 July 2024
SUPERSEDING
MIL-PRF-19500/735C
w/AMENDMENT 1
15 December 2022

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICES, DIODE, SILICON, SCHOTTKY,
DUAL DIODE, COMMON CATHODE, TYPE 1N7041
AND SINGLE DIODE TYPE 1N7045,
QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, Schottky, power rectifier diodes in both a dual diode common cathode and single diode configurations for use in high frequency switching applications. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type.

1.2 Package outlines. The device package outlines are as follows: A ceramic no lead chip carrier TO-276AB (U1 suffix) in accordance with figure 1 and three terminal flange mount header TO-257AA (T3 suffix) in accordance with figure 2.

1.3 Maximum ratings. Unless otherwise specified, maximum ratings at T_A = +25 °C.

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 5	Column 6
Types	V_{RWM}	I_O (1) (2) T_C = +100 °C	I_{FSM} (3) t_p = 8.3 ms, T_C = +25 °C	$R_{\theta JC}$ (4)	$R_{\theta JC}$	$R_{\theta JA}$	T_{STG} and T_J
	V dc	A dc	A (pk)	°C/W	°C/W	°C/W	°C
1N7041CCU1	45	20	250	1.67 (3)	0.83 (2)	80	-65 to +150
1N7045T3	45	10	110	2.6		80	

(1) See figures 3 and 4 for temperature-current derating curves.
(2) Entire package.
(3) Each leg.
(4) See figures 5 and 6 for thermal impedance curves.

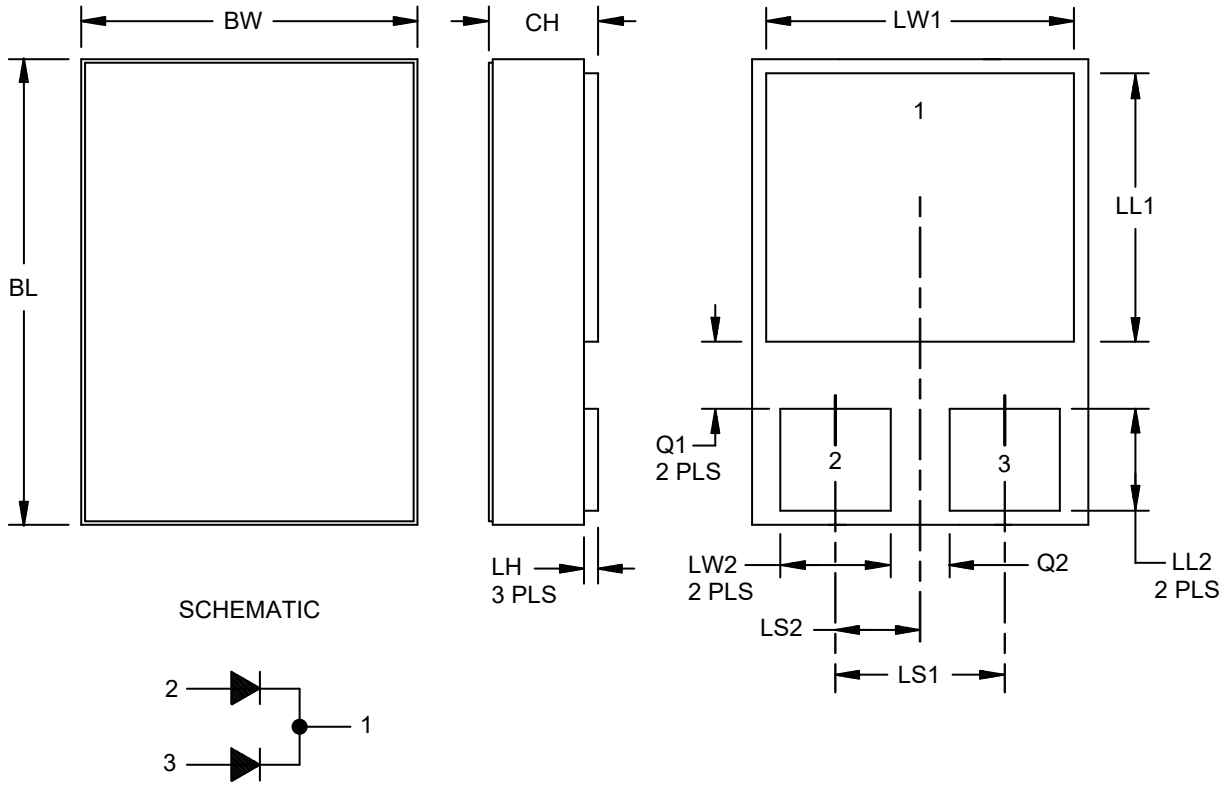
* 1.4 Primary electrical characteristics. There are no primary electrical characteristics listed for this specification sheet.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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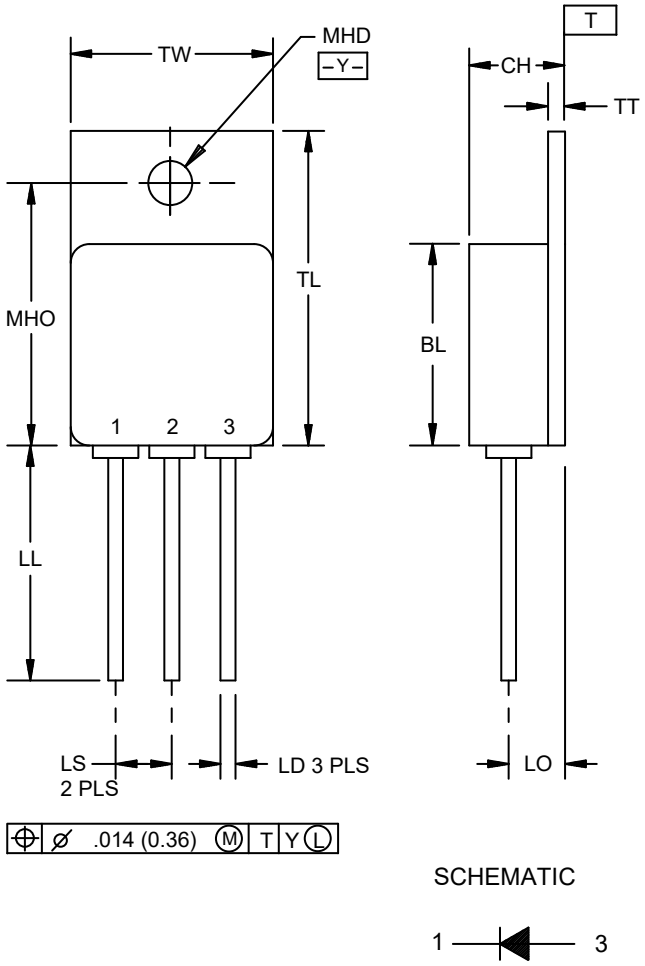


Symbol	Dimensions				Symbol	Dimensions			
	Inches	Inches	mm	mm		Inches	Inches	mm	mm
	Min	Max	Min	Max		Min	Max	Min	Max
BL	.620	.630	15.75	16.00	LS1	.200	.220	5.08	5.59
BW	.445	.455	11.30	11.56	LS2	.100	.110	2.54	2.79
CH	.129	.142	3.28	3.61	LW1	.370	.380	9.40	9.65
LH	.010	.020	0.25	0.51	LW2	.135	.145	3.43	3.68
LL1	.410	.420	10.41	10.67	Q1	.030		0.76	
LL2	.152	.162	3.86	4.11	Q2	.035		0.89	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Terminal 1 is common cathode. Terminal 2 is anode 1. Terminal 3 is anode 2.

FIGURE 1. Physical dimensions and configuration for TO-276AB package (1N7041CCU1).

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Symbol	Dimensions			
	Inches	Inches	mm	mm
	Min	Max	Min	Max
BL	.410	.430	10.41	10.92
CH	.190	.200	4.83	5.08
LD	.025	.035	0.64	0.88
LL	.500	.750	12.70	19.05
LO	.120 BSC		3.05 BSC	
LS	.100 BSC		2.54 BSC	
MHD	.140	.150	3.56	3.81
MHO	.527	.537	13.39	13.64
TL	.645	.665	16.38	16.89
TT	.035	.045	0.89	1.14
TW	.410	.420	10.41	10.67

SCHEMATIC

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Protrusion of ceramic eyelets included in dimension LL.
3. All terminals are isolated from case.

FIGURE 2. Dimensions and configuration for three terminal flange mount header TO-257AA package (1N7045T3).

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1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV" and "JANS".

1.5.2 Device type. The designation system for the device types of diodes covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The diodes of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the diodes covered by this specification sheet are "7041" and "7045".

1.5.3 Suffix symbols. The following suffix symbols apply to diodes covered by this specification sheet.

1.5.3.1 Schematic designator (1N7041 only). The suffix symbols "CC" designates that the internal schematic for the device is a dual anode, common cathode configuration.

1.5.3.2 Package suffix symbols. The following package designating suffix symbols are incorporated in the PIN for this specification sheet.

T3	Indicates a TO-257AA flange mount package in accordance with figure 2 (1N7045 only).
U1	Indicates a TO-276AB surface mount package (SMD-1) in accordance with figure 1 (1N7041 only).

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.
* MIL-STD-883 - Test Methods Standard Microcircuits.

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 and 2 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Lead formation. When lead formation is performed on the terminals of the TO-257 package, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14 of MIL-PRF-19500 and 100 percent dc testing in accordance with table I, subgroup 2 herein. Where a choice of lead formation is desired, it shall be specified in the acquisition document (see 6.2).

3.4.3 Lead isolation. Methods used for electrical isolation of the terminal feed through for the TO-257 package shall employ materials that contain a minimum of 90 percent ceramic AL_2O_3 or equivalent. Examples of such construction techniques are metalized ceramic eyelets or ceramic walled packages.

3.4.4 Polarity. Polarity and terminal configuration shall be in accordance with figures 1 and 2 herein.

* 3.4.5 Silicone die coating. The use of a silicone die coat requires a successful completion of method 5011 of MIL-STD-883 on each lot of silicone die coating for its intended applications.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500 and herein.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in tables I and II herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II herein).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E inspection. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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* 4.2.2 Silicone die coating. When silicone die coating is used, method 5011 of MIL-STD-883 shall be performed on that coating in the full MIL-PRF-19500 qualification process.

4.3 Screening (quality levels JANS, JANTXV, and JANTX only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANS level	Measurements for JANTX and JANTXV levels
3b	Surge current (see 4.3.4)	Surge current (see 4.3.4)
3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
3d	Avalanche energy test (see 4.3.3)	Avalanche energy test (see 4.3.3)
4	Required	Optional
5	Required Method 2052 of MIL-STD-750, PIND (see MIL-PRF-19500 and 4.3.5)	Not applicable
8	Required	Not applicable
9, 10	Not applicable	Not applicable
11	V_{F1} and I_{R1}	V_{F1} and I_{R1}
12	See 4.3.1, 240 hours	See 4.3.1, 48 hours
13	Subgroup 2 and 3, of table I herein, V_{F1} and I_{R1}; $\Delta V_{F1} \leq 50$ mV (pk); ΔI_{R1} = ±100 percent from the initial value or ±100 µA dc, whichever is greater.	Subgroup 2, of table I herein; V_{F1} and I_{R1}; $\Delta V_{F1} \leq 50$ mV (pk); ΔI_{R1} = ±100 percent from the initial value or ±100 µA dc, whichever is greater.

4.3.1 Power burn-in conditions. Power burn-in shall be performed in accordance with test condition A of method 1038 of MIL-STD-750. The following details shall apply: V_R = 36 V dc; T_J = +125 °C.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} and V_C. See table III, group E, subgroup 4 herein.

4.3.3 Avalanche energy test. The peak reverse energy test shall be performed in accordance with method 4064 of MIL-STD-750 using the circuit as shown on figure 7 or equivalent. The Schottky rectifier under test must be capable of absorbing the reverse energy, as follows: I_{AS} = 1 A, V_{BR} = 45 V minimum.

4.3.4 Surge current. The surge current test shall be performed in accordance with condition A of method 4066 of MIL-STD-750. The following details shall apply: I_O = 0; V_{RWM} = 0; I_{FSM} = rated I_{FSM} (see column 4 of 1.3); one pulse, tp = 8.3ms.

* 4.3.5 PIND. Not applicable in screening when devices are processed using alternative method and flow requirements approved by the qualifying activity, that includes incorporating the use of certified clean processing and silicone die coat. Instead, the PIND test performance shall be performed in group B3 and group C3, on a lot sample basis. PIND failures detected in group B or C will represent lot jeopardy and shall be evaluated for root cause and lot integrity.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500, and table I herein. Electrical measurements (end-points) and delta measurements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables E-VIA (for quality level JANS) and E-VIB (for quality levels JAN, JANTX, and JANTXV) of MIL-PRF-19500 and as follows. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

	Subgroup	Method	Condition
*	B3	2052	PIND, required if not performed in screening. (22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
	B4	1037	ΔT_C = +85 °C, I_F = 2 A minimum.
	B5	1038	Condition A, V_R = 36 V dc, T_J = +125 °C, t = 340 hours min; heat sinking allowed. This test shall be extended to 1000 hours on each JANS wafer lot.

4.4.2.2 Quality levels JAN, JANTX, and JANTXV, table E-VIB of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1037	ΔT_C = +85 °C, I_F = 2 A minimum.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. Delta measurements shall be in accordance with table II herein.

	Subgroup	Method	Condition
	C2	2036	Condition A, weight = 10 lbs (4.54 kg), t = 15 seconds. Not required for 1N7041.
*	C3	2052	PIND, required if not performed in screening. (JANS only, 22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
	C5	4081	Limit for thermal resistance for 1N7041 is 1.67 °C/W for each leg. Limit for thermal resistance for 1N7045 is 2.6 °C/W.
	C6	1037	ΔT_C = +85 °C, I_F = 2 A minimum.
	C6	1038	Condition A, V_R = 36 V dc, T_J = +125 °C, t = 1,000 hours minimum; (heat sinking allowed). (required for quality levels TX and TXV only)

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-IX of MIL-PRF-19500, and table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein except $Z_{\theta JX}$ need not be performed after group E subgroup 2.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. The conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 3/	3101	See 4.3.2	$Z_{\theta JC}$			°C/W
Forward voltage	4011	Condition B, pulsed (see 4.5.1);	V_{F1}			
1N7041		I_F = 10 A (pk)			0.57	V dc
1N7045		I_F = 10 A (pk)			0.73	V dc
Forward voltage	4011	Condition B, pulsed (see 4.5.1);	V_{F2}			
1N7041		I_F = 20 A (pk)			0.69	V dc
1N7045		I_F = 20 A (pk)			1.02	V dc
Reverse current	4016	Condition A or B;	I_{R1}			
1N7041		V_R = 45 V			0.5	mA dc
1N7045		V_R = 45 V			0.5	mA dc
Subgroup 3						
High temperature operation:		T_C = +125 °C				
Forward voltage	4011	Condition B, pulsed (see 4.5.1);	V_{F3}			
1N7041		I_F = 10 A (pk)			0.48	V dc
1N7045		I_F = 10 A (pk)			0.70	V dc
Forward voltage	4011	Condition B, pulsed (see 4.5.1);	V_{F4}			
1N7041		I_F = 20 A (pk)			0.61	V dc
1N7045		I_F = 20 A (pk)			1.10	V dc
Reverse current	4016	Condition A or B;	I_{R2}			
1N7041		V_R = 45 V			67	mA dc
1N7045		V_R = 45 V			67	mA dc
Low temperature operation:		T_C = -55 °C				
Forward voltage	4011	Condition B, pulsed (see 4.5.1);	V_{F5}			
1N7041		I_F = 10 A dc			0.65	V dc
1N7045		I_F = 10 A dc			0.74	V dc
Forward voltage	4011	Condition B, pulsed (see 4.5.1);	V_{F6}			
1N7041		I_F = 20 A dc			0.74	V dc
1N7045		I_F = 20 A dc			0.95	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Junction capacitance	4001	V_R = 5 V dc, f = 1 MHz, V_{SIG} = 50 mV (p-p)	C_J		900	pF
Subgroup 5						
Not applicable						
Subgroup 6						
Surge current	4066	Condition A, I_{FSM} = rated I_{FSM} (see column 4 of 1.3). Ten surges for each internal diode. 60 seconds between surges, (see 4.5.1)				
Electrical measurements		See table I, subgroup 2 herein				
Subgroup 7						
Dielectric withstanding voltage	1081	V_R = 500 V dc; all leads shorted, measure from leads to case	I_{DWV}		10	µA dc
Scope display evaluation	4023	Condition A (stable only)				
Electrical measurements		See table I, subgroup 2 herein				

1/ For sampling plan, see MIL-PRF-19500.
2/ Electrical characteristics apply to all package styles and polarities.
3/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 3 and 4 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, JANTXV).
Group C, subgroups 2 and 6.
Group E, subgroup 1.

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TABLE II. Delta measurements for groups B, C, and E. 1/ 2/ 3/ 4/ 5/ 6/

Step	Inspection	MIL-STD-750		Symbol	Limits	Limits	Unit
		Method	Conditions		Min	Max	
1	Forward voltage 1N7041 1N7045	4011	Condition B; I_F = 10 A (pk) pulsed (see 4.5.1) I_F = 10 A (pk) pulsed (see 4.5.1)	ΔV_{F1}	±50 mV dc from initial reading.		
2	Reverse current	4016	Condition A or B; V_R = 45 V	ΔI_{R1}	±100 percent from the initial value or ±100µA dc, whichever is greater.		

1/ Each internal diode.
2/ The delta measurements taken during group B inspection for quality level JANS (table E-VIA of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 4, the measurements of steps 1, and 2 shall also be taken.
   b. In addition to the measurements specified for subgroup 5, the measurements of steps 1 and 2 shall also be taken.
3/ The delta measurements taken during group B inspection for quality levels JANTX and JANTXV (table E-VIB of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 2, the measurements of steps 1, and 2 shall also be taken.
   b. In addition to the measurements specified for subgroup 3, the measurements of steps 1, and 2 shall also be taken.
   c. In addition to the measurements specified for subgroup 6, the measurements of steps 1 and 2 shall also be taken.
4/ The delta measurements taken during group C inspection for all quality levels (table E-VII of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroups 2 and 3, the measurements of steps 1, and 2 shall also be taken.
   b. In addition to the measurements specified for subgroup 6, the measurements of steps 1, 2 shall also be taken.
5/ The delta measurements taken during E inspection for all quality levels (table E-IX of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 1, the measurements of steps 1, and 2 shall also be taken.
   b. In addition to the measurements specified for subgroup 2, the measurements of steps 1 and 2 shall also be taken.
6/ Devices which exceed the table I limits for this test shall not be accepted.

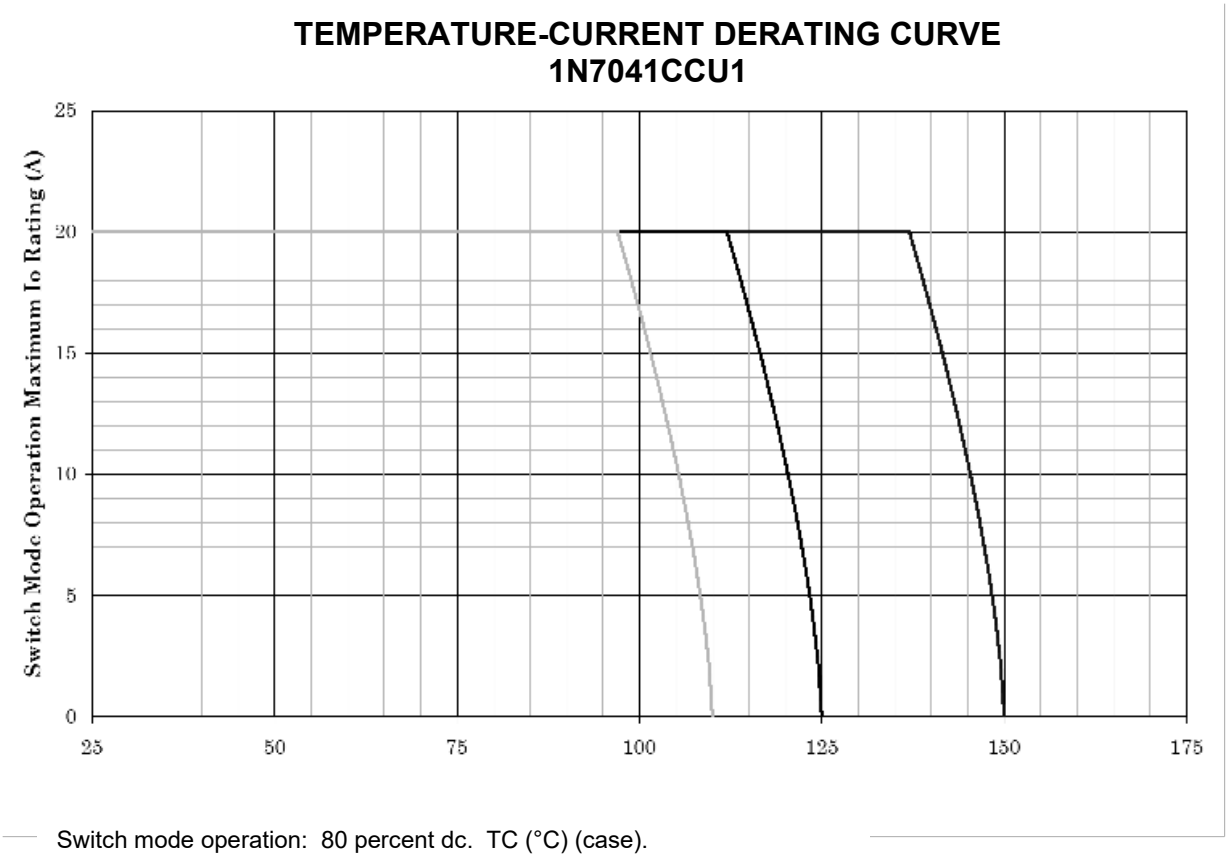
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TABLE III. Group E inspection (all quality levels) – for qualification and requalification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Temperature cycling (air to air)	1051	Test condition G, 500 cycles, −55 °C to +150 °C.	
Hermetic seal	1071	Fine and gross leak.	
Electrical measurements		See table I, subgroup 2.	
Subgroup 2			45 devices, c = 0
Life test	1048	t = 1,000 hours, T_J = +125 °C, V_R = 80 percent rated voltage (see 1.3, column 2 herein).	
Electrical measurements		See table I subgroup 2.	
Subgroup 4			
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 10 1/			5 devices, c = 0
Surge current 1N7041	4066	Condition A, T_A = +25 °C, I_{FSM} = 250 A, 100 surges of 8.3 ms superimposed on I_O. V_R = 0; I_O = 0 A pk half sine wave.	
1N7045		Condition A, T_A = +25°C, I_{FSM} = 110 A, 100 surges of 8.3 ms superimposed on I_O. V_R = 0; I_O = 0 A pk half sine wave.	
Electrical measurements		See table I subgroup 2 (V_F and I_R only).	

1/ Each individual diode.

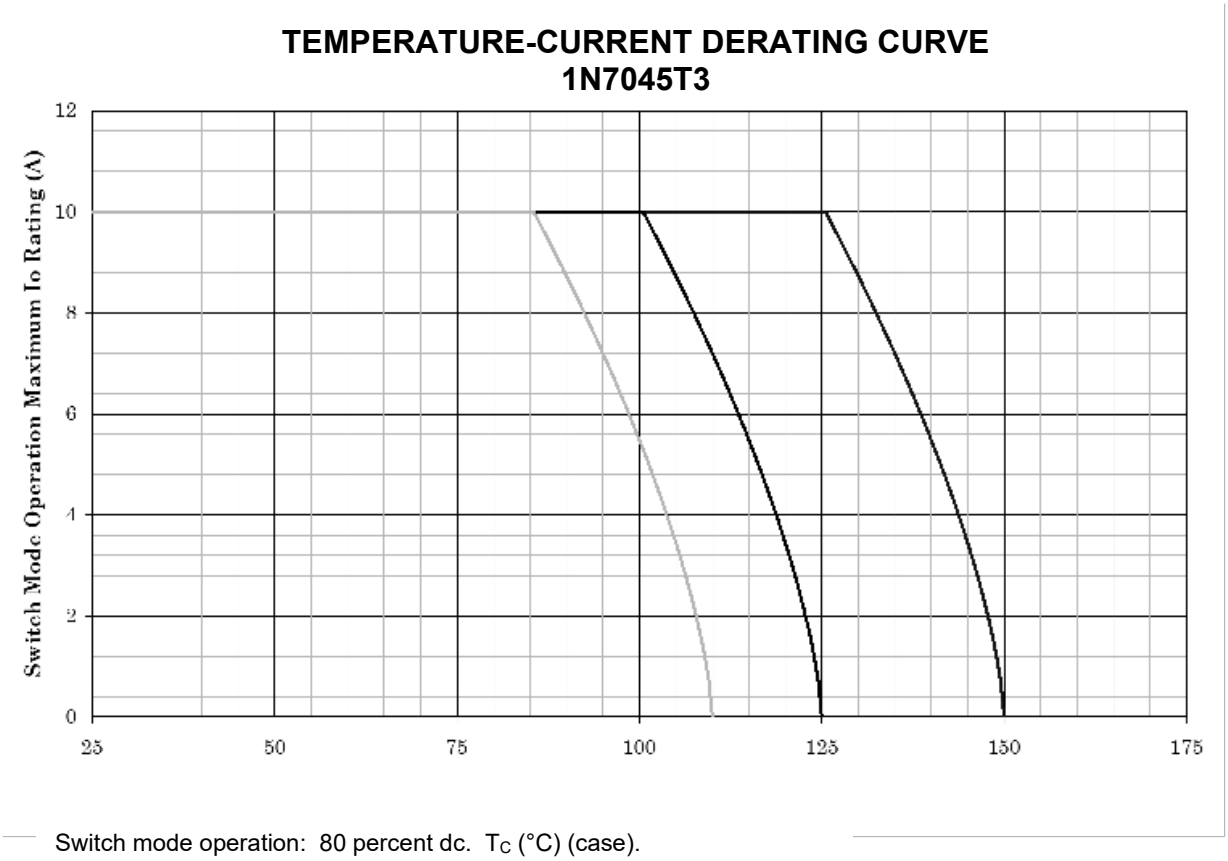
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Switch mode operation: 80 percent dc. TC (°C) (case).
Thermal resistance junction to case = 0.83°C/W.

FIGURE 3. Temperature-current derating curve (entire package) 1N7041CCU1.

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Switch mode operation: 80 percent dc. T_C (°C) (case).
Thermal resistance junction to case = 2.6 °C/W.

FIGURE 4. Temperature-current derating curve for 1N7045T3.

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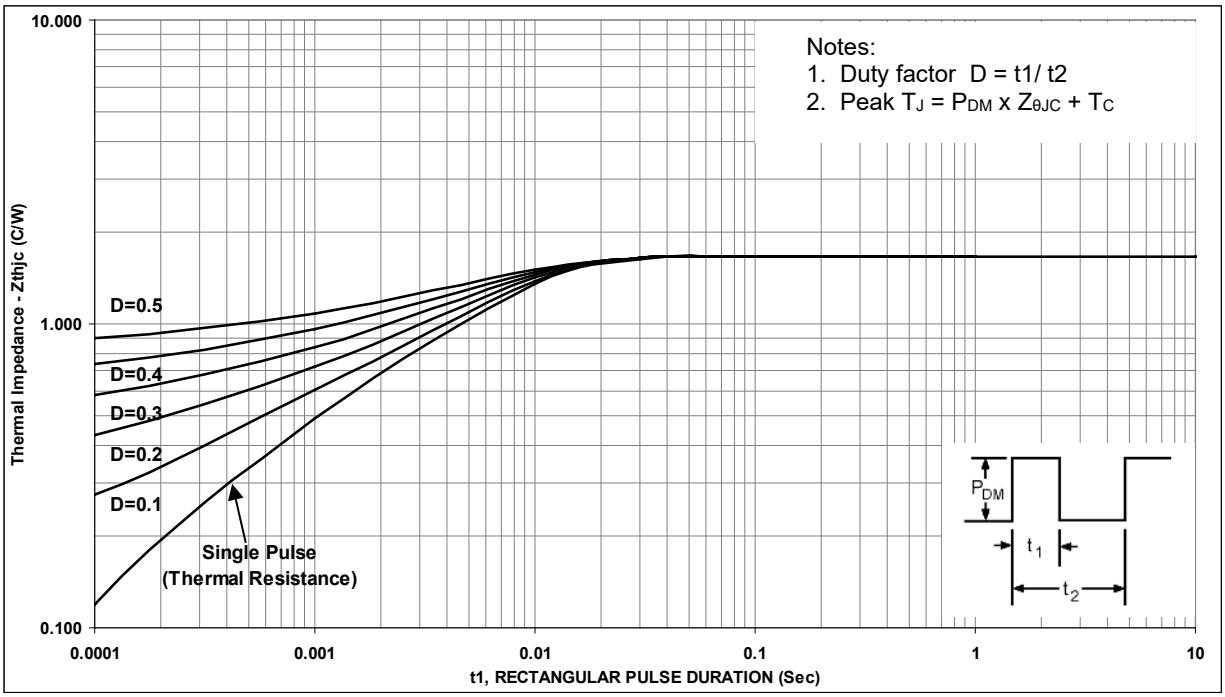


FIGURE 5. Thermal impedance (for each leg) 1N7041CCU1.

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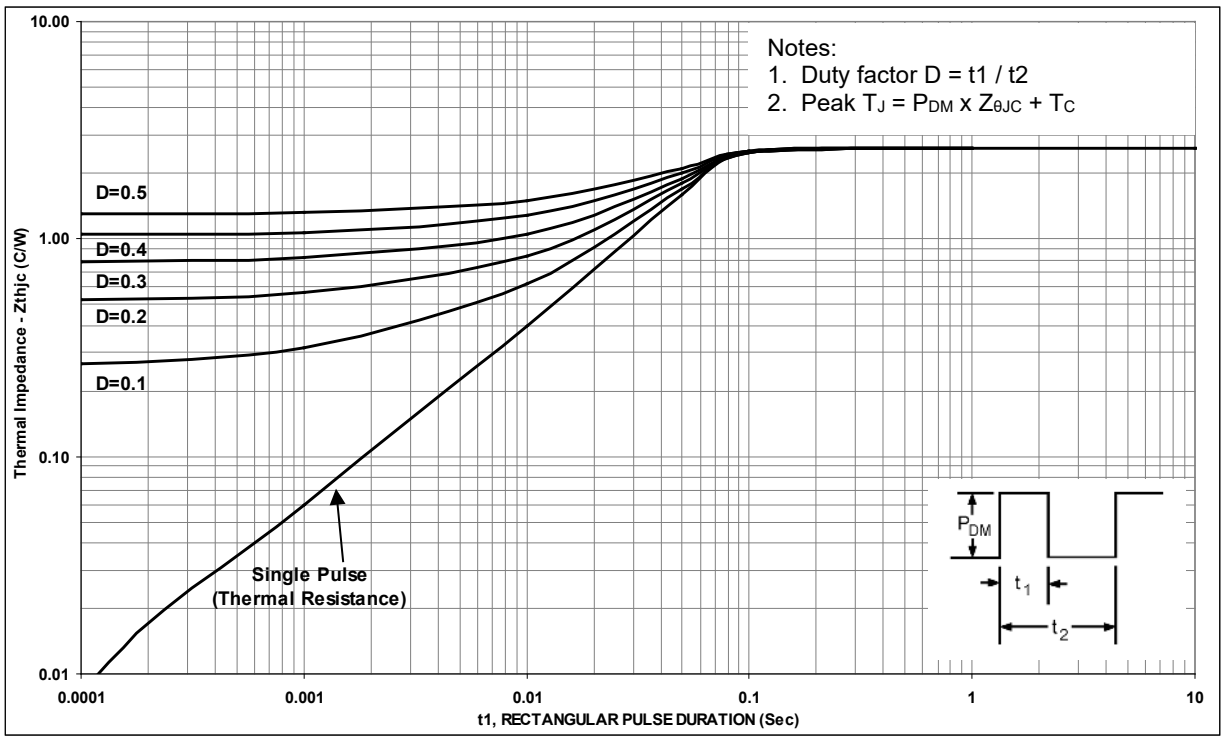
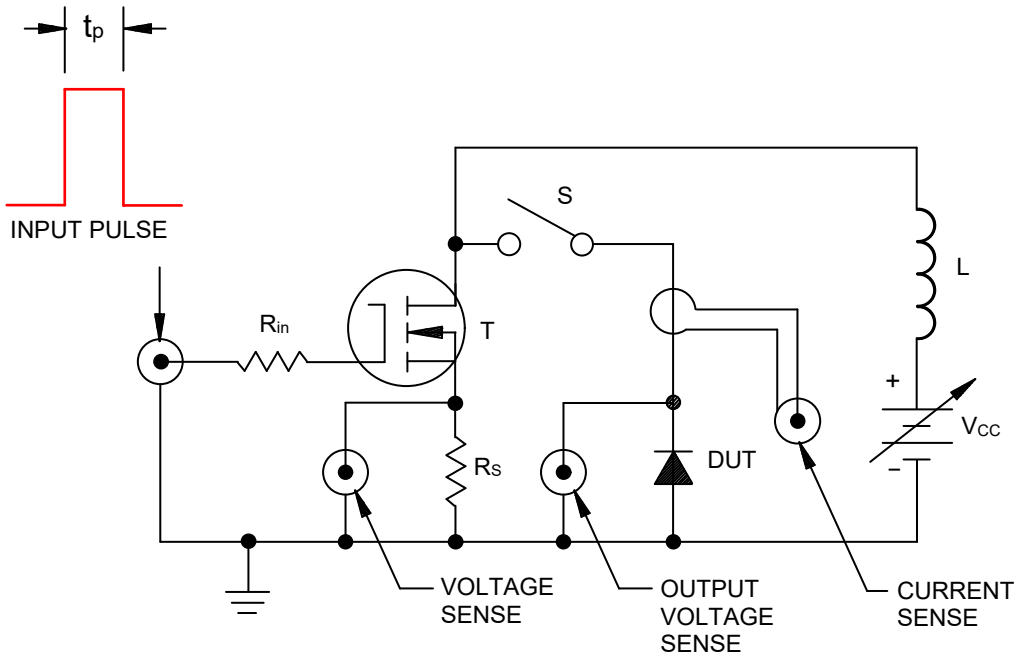


FIGURE 6. Thermal impedance for 1N7045T3.

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PROCEDURES:
1. With S open, adjust pulse width to test current of 1 amps across R_S.
2. Close S, verify test current with current sense.
3. Read peak output voltage (see 4.3.3).

NOTES:
1. The following input pulse details shall apply: V_G = 10 Volts, duty cycle ≤ 1 percent.
2. The following test circuit component values shall apply: L = 100 µH, Z_G = 50 ohms, R_{in} = 50 ohms, 1 watt, R_S = 0.1 ohms, 1 watt, T = 2N6764 (IRF150 or equivalent).
3. The supply voltage V_{CC} ≈ 10 volts.

FIGURE 7. Avalanche energy test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

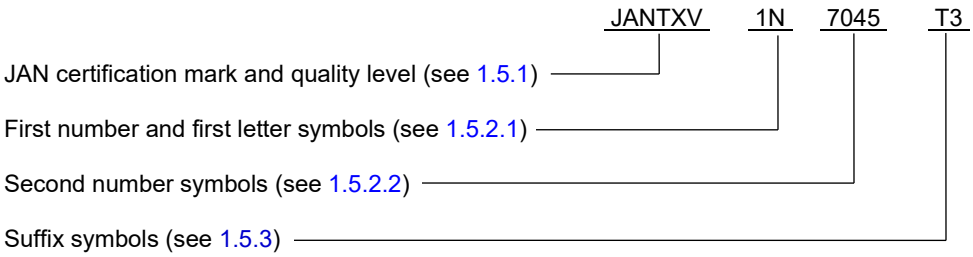
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1) and if applicable, any needed lead formation (see 3.4.2).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



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6.5 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
JAN1N7041CCU1	JANTX1N7041CCU1	JANTXV1N7041CCU1	JANS1N7041CCU1
JAN1N7045T3	JANTX1N7045T3	JANTXV1N7045T3	JANS1N7045T3

6.6 Cross reference substitution list. A PIN for PIN replacement table follows, and these devices are directly interchangeable.

Non-preferred PIN	Preferred PIN
20CLQ045	JANS, JANTXV, JANTX, JAN1N7041CCU1
10YQ045C	JANS, JANTXV, JANTX, JAN1N7045T3

6.7 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army − CR
Navy - SH
Air Force − 85
NASA − NA
DLA − CC

Preparing activity:
DLA − CC

(Project 5961-2024-059)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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